HEALTH AI

Co-Creation & Innovation Platform

Software Requirements Specification

(SRS)

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16/04/2026	1.0	Initial version	System

Table of Contents

Reading Guide: v1 Scope vs. Phase 2 Target

This document describes the complete requirements of the HEALTH AI Co-Creation & Innovation Platform. The platform is currently in its v1 release stage, and the full scope defined in the project brief will be delivered incrementally.

To help the reader distinguish between what is currently implemented and what is planned for a future release, every functional requirement, non-functional requirement, use case, and data entity is tagged with a phase label:

[v1]	Currently implemented in the codebase and available in the production build.
[Phase 2]	Defined in the project brief as a mandatory requirement; planned for a future release and not yet implemented.

Requirements without an explicit phase label inherit the phase indicated in the Phase column of the corresponding requirements table.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the HEALTH AI Co-Creation & Innovation Platform, internally known as "Project Bridge." The document defines the functional and non-functional requirements, system architecture, data model, interface specifications, and use case scenarios of the platform. It serves as the primary reference for the development team, project stakeholders, quality assurance personnel, and course instructors.

1.2 Scope

The HEALTH AI platform is a structured web-based collaboration system that enables Engineers and Healthcare Professionals to discover compatible partners, form trusted connections, and coordinate cross-disciplinary projects. The platform supports partner discovery through structured posts, interest expression with NDA-protected confidentiality, a meeting request workflow culminating in externally-hosted meetings, an Administrative Dashboard for oversight, and a complete activity audit trail.

The platform does not provide financial transactions, contract management, medical advice, video conferencing, file upload/sharing capabilities, or the storage of patient medical records. Meetings themselves are conducted on external platforms such as Zoom or Microsoft Teams; the platform only facilitates scheduling and context-setting.

1.3 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification
RBAC	Role-Based Access Control
GDPR	General Data Protection Regulation
NDA	Non-Disclosure Agreement
WCAG	Web Content Accessibility Guidelines
API	Application Programming Interface
JSON	JavaScript Object Notation
SPA	Single Page Application
XSS	Cross-Site Scripting
ARIA	Accessible Rich Internet Applications
scrypt	A password-based key derivation function used for password hashing
Post	A structured collaboration opportunity published by an Engineer or Healthcare Professional
Focused Room	A post-attached conversation thread; the v1 implementation of the partner communication channel
Meeting Request	The formal workflow through which an interested party proposes a meeting with a post owner
Partner Found	The terminal state of a successful post lifecycle indicating a collaborator has been secured

1.4 Intended Audience

This document is intended for the software development team responsible for building and maintaining the platform, the course instructor and teaching assistants for evaluation purposes, quality assurance testers who will verify the system against these requirements, and any stakeholders who need to understand the platform's capabilities, current v1 scope, and planned Phase 2 features.

2. Overall Description

2.1 Product Perspective

The HEALTH AI platform addresses a concrete gap in how healthcare professionals and engineers discover collaboration opportunities. Traditional channels such as LinkedIn, email outreach, and conference networking rely on coincidence and lack project-specific context, making it difficult for a doctor with a clinical problem to find an engineer with the right technical skills (or vice versa).

The platform creates a purposeful environment where ideas are visible as structured posts, expertise is discoverable through filters and search, confidentiality is protected through NDAs, and interactions progress through a defined workflow from interest expression through meeting scheduling to partner confirmation. An administrative layer ensures content quality and full auditability.

2.2 System Architecture

The system follows a client-server architecture. The frontend is a Single Page Application (SPA) served as static files (HTML, CSS, JavaScript) that communicates with a REST API backend. The backend is a Node.js HTTP server that handles API requests and serves static assets. In v1, data storage uses a JSON file with atomic write operations managed through a sequential Promise-based queue. The application supports deployment both locally and on Vercel's serverless platform. Phase 2 will migrate persistent storage to a relational database to support activity logging, RBAC, and higher concurrent user volumes.

2.3 User Roles

Role	Description	Key Permissions
Engineer	A technical professional (software engineer, hardware engineer, data scientist, designer, etc.) who creates posts describing technical solutions or capabilities and seeks healthcare partners to apply them to real clinical problems.	Register with .edu email [Phase 2], login, create/edit own posts, browse network, express interest, accept NDA, propose meeting slots, mark Partner Found on own posts, manage profile
Healthcare Professional	A clinician, researcher, or healthcare institution member who creates posts describing clinical needs or research problems and seeks engineering partners to co-develop solutions.	Register with .edu email [Phase 2], login, create/edit own posts, browse network, express interest, accept NDA, propose meeting slots, mark Partner Found on own posts, manage profile
Admin [Phase 2]	A platform moderator with elevated privileges for content oversight, user management, and audit trail access. Not available in v1.	All user permissions PLUS view/remove any post, view/suspend any user, view full activity logs, export logs as CSV, view platform statistics

v1 IMPLEMENTATION NOTE In v1, all authenticated users have identical permissions regardless of their selected profession. Role-Based Access Control (RBAC) and the Admin role will be introduced in Phase 2.

2.4 Assumptions and Dependencies

It is assumed that users have access to a modern web browser (Chrome, Firefox, Edge, or Safari, latest versions) and a stable internet connection. In Phase 2, it is assumed that users have an institutional .edu or .edu.tr email address for registration. Node.js runtime is required on the server. The JSON file-based storage is suitable for the v1 scale; Phase 2 will introduce a database-backed store. On Vercel deployments, data stored in /tmp is ephemeral and is not suitable for production; this constraint will be resolved in Phase 2. The platform relies on no external services or third-party libraries on the server side in v1.

2.5 Constraints

The platform shall not store patient data or sensitive medical records. File uploads are not supported in any phase. Meetings take place on external platforms; the platform only facilitates context-setting and scheduling. Data persistence on Vercel is temporary due to serverless architecture limitations. In v1, there is no admin panel, no email verification, no RBAC enforcement, and no activity audit trail; these are Phase 2 deliverables. In v1, the post lifecycle state model is not yet enforced at the backend.

3. Functional Requirements

Each functional requirement is defined with a unique identifier, description, priority level (High/Medium/Low), release phase (v1 or Phase 2), and source reference to the project brief section or implementing code component.

3.1 User Registration & Access Control

The platform supports a two-role registration system (Engineer and Healthcare Professional) with institutional email verification. An Admin role is created through an internal provisioning process in Phase 2 and is not self-registerable.

ID	Requirement Description	Priority	Phase	Source
FR-01	The system shall only allow registration with institutional email addresses (.edu or .edu.tr extensions).	High	**Phase 2**	Brief 4.1
FR-02	The system shall implement an email verification mechanism requiring users to confirm their address via a confirmation link before account activation.	High	**Phase 2**	Brief 4.1
FR-03	The user shall be able to select a role during registration: Engineer, Healthcare Professional, or Admin.	High	**Phase 2**	Brief 4.1
FR-04	The system shall allow user registration with email, first name, last name, phone, password, and profession.	High	**v1**	server.js /api/register
FR-05	The system shall validate that the email address follows a valid format (user@domain.tld).	High	**v1**	server.js isValidEmail()
FR-06	The system shall validate that the phone number contains at least 10 digits.	High	**v1**	server.js isValidPhone()
FR-07	The system shall enforce a minimum password length of 8 characters.	High	**v1**	server.js /api/register
FR-08	The system shall hash passwords using scrypt with a 16-byte random salt before storage.	High	**v1**	server.js hashPassword()
FR-09	The system shall prevent duplicate registrations by checking existing emails case-insensitively.	High	**v1**	server.js /api/register
FR-10	The system shall support login via email and password with timing-safe password comparison.	High	**v1**	server.js /api/login
FR-11	The system shall return a session object containing userId and email upon successful login.	High	**v1**	server.js /api/login
FR-12	The system shall store session data in the browser's sessionStorage during the browser session.	Medium	**v1**	app.js auth layer
FR-13	The system shall provide a logout function that clears session data and returns the user to the login screen.	High	**v1**	app.js logout()
FR-14	The system shall enforce Role-Based Access Control (RBAC) restricting Admin-only actions to users with the Admin role.	High	**Phase 2**	Brief 4.5
FR-15	The system shall provide password reset via email link for users who have forgotten their credentials.	Medium	**Phase 2**	Brief 4.1

PHASE 2 TARGET FR-01, FR-02, FR-03, FR-14, and FR-15 capture the brief's access-control model (institutional .edu emails, email verification, two-role self-registration, RBAC, password reset). v1 currently accepts any valid email address and offers free-form profession selection without role-based permissions.

3.2 Post Management

A Post represents a collaboration opportunity published by an Engineer or a Healthcare Professional. Each post follows a defined lifecycle: Draft → Active → Meeting Scheduled → Partner Found, with an automatic Expired path for inactive posts. Posts include a confidentiality level that determines whether NDA acceptance is required before interaction.

ID	Requirement Description	Priority	Phase	Source
FR-20	The system shall allow Engineers and Healthcare Professionals to create posts describing collaboration opportunities.	High	v1	server.js /api/projects
FR-21	A post shall contain: title, working domain, required expertise, project stage, confidentiality level, city, and description.	High	Phase 2	Brief 4.2 / UserGuide 3.2
FR-22	The system shall support the following post lifecycle states: Draft, Active, Meeting Scheduled, Partner Found, and Expired.	High	Phase 2	Brief 4.2
FR-23	A newly created post shall enter the Draft state if saved without publishing, or Active state if published immediately.	High	Phase 2	Brief 4.2
FR-24	A post shall automatically transition to Meeting Scheduled state when a meeting is confirmed between the owner and an interested party.	High	Phase 2	Brief 4.2
FR-25	A post shall transition to Partner Found state when the owner explicitly marks it as such; in this state the post no longer accepts new interest.	High	Phase 2	Brief 4.2 / UserGuide 3.5
FR-26	A post shall automatically transition to Expired state after a configurable inactivity period (default: 60 days).	Medium	Phase 2	Brief 4.2
FR-27	The system shall validate that title, domain, summary, collaborationGoal, and ownerId are provided when creating a post.	High	v1	server.js /api/projects
FR-28	The system shall automatically assign the logged-in user as the post owner.	High	v1	app.js projectForm handler
FR-29	The system shall allow post owners to edit their posts while in Draft or Active state.	High	Phase 2	UserGuide 3.3
FR-30	The system shall display a visual status badge (Gray: Draft, Green: Active, Blue: Meeting Scheduled, Purple: Partner Found, Red: Expired).	Medium	Phase 2	UserGuide 3.4
FR-31	The system shall support a Confidentiality Level field with values: Public or Confidential.	Medium	Phase 2	Brief 4.2 / UserGuide 3.2
FR-32	The system shall generate a unique post ID with a 'p' prefix for each new post.	High	v1	server.js createId()
FR-33	The system shall display all posts in reverse chronological order.	Medium	v1	app.js loadBootstrap()

FR-34	The system shall support a supplementary project stage classification (Discovery, Prototype, Validation, Launch Prep) within Active posts.	Low	**v1**	app.js project form

> **v1 IMPLEMENTATION NOTE** In v1, posts correspond to the 'projects' entity in the codebase. The stage field (Discovery, Prototype, Validation, Launch Prep) captures project maturity as a supplementary dimension and does not replace the post lifecycle model (Draft/Active/Meeting Scheduled/Partner Found/Expired), which is tracked under FR-22 as a Phase 2 deliverable.

3.3 Search & Matching

Users shall be able to discover relevant posts and potential partners through filtering and free-text search. City-based matching is a first-class capability intended to bring collaborators together within the same geographic area.

ID	Requirement Description	Priority	Phase	Source
FR-40	The system shall allow authenticated users to browse and search all active posts.	High	**v1**	app.js renderProjects()
FR-41	The system shall allow filtering posts by domain, required expertise, status, and city.	High	**Phase 2**	Brief 4.3 / UserGuide 4.2
FR-42	The system shall support city-based matching to prioritize partners in the same geographic area.	High	**Phase 2**	Brief 4.3 / UserGuide 4.3
FR-43	The system shall provide real-time text search filtering users by name, role, specialty, city, status, bio, availability, and interests.	High	**v1**	app.js peopleSearch handler
FR-44	The system shall display user cards with name, specialty, city, role, bio, status, availability, and interests on the Network page.	Medium	**v1**	app.js renderPeople()
FR-45	The system shall display availability badges derived from user status and availability text.	Medium	**v1**	app.js getAvailabilityBadge()

3.4 Meeting Request Workflow

The meeting request workflow is the primary collaboration channel defined in the project brief. The flow begins with an interest expression on a post, proceeds through NDA acceptance (if the post is Confidential), continues with time-slot proposal and acceptance, and concludes with an externally-hosted meeting whose outcome may transition the post to Partner Found.

ID	Requirement Description	Priority	Phase	Source
FR-50	The system shall allow an interested party to express interest in a post via 'Express Interest' / 'Request Meeting'.	High	**Phase 2**	Brief 4.4 / UserGuide 5.1
FR-51	When interest is expressed on a Confidential post, the system shall require NDA acceptance before the request proceeds.	High	**Phase 2**	Brief 4.4 / UserGuide 5.2
FR-52	The system shall record NDA acceptance with a timestamp associated with the user and the post.	High	**Phase 2**	Brief 4.4
FR-53	The interested party shall be able to propose one or more available time slots for the meeting.	High	**Phase 2**	Brief 4.4 / UserGuide 5.3

FR-54	The post owner shall be able to review proposed slots and accept one; upon acceptance the meeting is confirmed.	High	**Phase 2**	Brief 4.4 / UserGuide 5.3
FR-55	Upon meeting confirmation, both parties shall receive a notification containing the agreed time and a placeholder for the external meeting link (Zoom/Teams).	High	**Phase 2**	Brief 4.4 / UserGuide 5.3
FR-56	The system shall not host meetings itself; meetings take place on external platforms and only the context/link is shared.	High	**Phase 2**	Brief 4.4 / UserGuide 5.3
FR-57	After a confirmed meeting, the post owner shall be able to mark the post as Partner Found or leave it Active.	High	**Phase 2**	UserGuide 5.4
FR-58	As an interim v1 alternative to the meeting workflow, the system shall provide a Focused Room where both parties exchange messages attached to a post context.	High	**v1**	server.js /api/conversations
FR-59	The system shall allow participants to send text messages within a Focused Room with timestamped delivery.	High	**v1**	server.js /api/messages
FR-60	The system shall validate that the message sender is a participant in the conversation before accepting the message.	High	**v1**	server.js /api/messages
FR-61	The system shall prevent duplicate Focused Rooms between the same participants for the same post.	High	**v1**	server.js /api/conversations

v1 IMPLEMENTATION NOTE In v1, the full meeting request workflow (interest, NDA, slot proposal, scheduling) has not been implemented. As an interim solution, FR-58 through FR-61 deliver a Focused Room capability: when a user shows interest in a post, a project-attached conversation room is opened directly and the two parties exchange messages. This preserves the spirit of "contextual partner communication" while the full workflow is delivered in Phase 2, at which point Focused Rooms will become the message history transcript for a scheduled meeting rather than a replacement for it.

3.5 Administrative Dashboard

The Administrative Dashboard is a Phase 2 deliverable that gives Admins moderation tools for posts and users and visibility into platform-wide usage metrics.

ID	Requirement Description	Priority	Phase	Source
FR-70	The system shall provide an Administrative Dashboard accessible only to users with the Admin role.	High	**Phase 2**	Brief 4.5
FR-71	Admins shall be able to view, filter, and remove inappropriate posts.	High	**Phase 2**	Brief 4.5 / UserGuide 7.1
FR-72	Admins shall be able to list, filter, and view user profiles.	High	**Phase 2**	Brief 4.5 / UserGuide 7.2
FR-73	Admins shall be able to suspend user accounts; suspended accounts cannot log in until reactivated.	High	**Phase 2**	Brief 4.5 / UserGuide 7.2

FR-74	Admins shall be able to view platform-wide statistics (users per role, posts per status, meetings scheduled, partners found).	Medium	**Phase 2**	Brief 4.5
FR-75	Admins shall be able to export activity logs as CSV files.	Medium	**Phase 2**	UserGuide 7.3

> **PHASE 2 TARGET** The entire Administrative Dashboard is scheduled for Phase 2. In v1, there is no Admin role and no admin UI; moderation actions, if required, are performed through direct database edits.

3.6 Activity Logging & Audit Trail

Activity logging provides an immutable audit trail of all significant user and admin actions. This is essential for GDPR compliance, security incident investigation, and administrative oversight.

ID	Requirement Description	Priority	Phase	Source
FR-80	The system shall maintain an activity log for every significant action: registration, login, post lifecycle transitions, interest, NDA, meeting events, partner-found, and admin actions.	High	**Phase 2**	Brief 4.6
FR-81	Each log entry shall contain: timestamp, userId, action type, target entity, IP address, and outcome.	High	**Phase 2**	Brief 4.6
FR-82	Activity logs shall be immutable; no user, including Admin, shall be able to edit or delete individual entries.	High	**Phase 2**	Brief 4.6
FR-83	Admins shall be able to filter logs by date range, user, and action type.	Medium	**Phase 2**	UserGuide 7.3
FR-84	The system shall retain activity logs for a minimum of 12 months.	Medium	**Phase 2**	Brief 4.6

> **PHASE 2 TARGET** Activity logging is entirely a Phase 2 deliverable. v1 does not record persistent audit events; the only record of user actions is the data produced by those actions (created posts, messages, connections).

3.7 Profile Management

ID	Requirement Description	Priority	Phase	Source
FR-90	The system shall display a detailed profile page for each user (name, role, specialty, city, bio, status, availability, interests, connections, posts).	High	**v1**	app.js renderProfileSummary()
FR-91	The system shall allow users to edit their own profile information.	High	**Phase 2**	UserGuide 6.1
FR-92	The system shall allow users to permanently delete their own account (GDPR right to erasure).	High	**Phase 2**	UserGuide 6.2
FR-93	The system shall allow users to export their personal data in a downloadable format (GDPR data portability).	Medium	**Phase 2**	UserGuide 6.3

FR-94	The system shall allow users to add other users to their network (bidirectional connection).	Medium	**v1**	server.js /api/friends
FR-95	The system shall allow browsing any member's profile without switching the signed-in account.	Medium	**v1**	app.js profile selector

3.8 User Interface, Navigation & Notifications

ID	Requirement Description	Priority	Phase	Source
FR-100	The system shall provide three visual themes (Neon Pulse, Earth Canvas, Citrus Tide) cycleable via a toggle.	Low	**v1**	app.js setTheme()
FR-101	The system shall provide main navigation sections: Overview, Project Board, Network, Create, Focused Contact, Rooms Hub, Member Profiles, and Administrative Dashboard (Admin only).	High	**v1 / Phase 2**	app.js navigation
FR-102	The system shall support hash-based routing with browser history navigation.	Medium	**v1**	app.js hashchange
FR-103	The system shall display a dashboard with key statistics on the Overview page.	Medium	**v1**	app.js renderStats()
FR-104	The system shall display toast notifications for user actions.	Low	**v1**	app.js showNotice()
FR-105	The system shall deliver in-app notifications for: interest received, meeting request, meeting confirmed, post status changed, account activity.	High	**Phase 2**	UserGuide 8
FR-106	The system shall deliver email notifications for: account verification, password reset, meeting confirmation, and security-relevant account activity.	High	**Phase 2**	UserGuide 8

4. Non-Functional Requirements

The following non-functional requirements define the quality attributes, performance targets, and compliance constraints of the system across both phases.

ID	Requirement	Metric / Target	Category	Phase
NFR-01	Search results shall be returned within a specified time limit.	< 1.5 seconds	Performance	**v1**
NFR-02	The initial page load shall complete within a reasonable threshold.	< 3 seconds	Performance	**v1**
NFR-03	Client-side search filtering shall produce results within a responsive threshold.	< 200 ms	Performance	**v1**
NFR-04	The system shall support concurrent authenticated users without degradation.	>= 500 concurrent	Performance	**Phase 2**
NFR-05	Passwords shall be hashed using scrypt with 16-byte salt and 64-byte key length.	scrypt(salt=16B, keylen=64B)	Security	**v1**
NFR-06	Password comparison shall use timing-safe comparison.	crypto.timingSafeEqual	Security	**v1**
NFR-07	All user input shall be sanitized before rendering or storage.	XSS prevention	Security	**v1**
NFR-08	Request body size shall be limited to prevent denial-of-service.	Max 1 MB payload	Security	**v1**
NFR-09	The system shall enforce HTTPS in production.	TLS 1.2+	Security	**Phase 2**
NFR-10	Session tokens shall expire after defined inactivity.	Expire after 24h idle	Security	**Phase 2**
NFR-11	The system shall comply with GDPR (consent, access, erasure, portability).	GDPR full compliance	GDPR / Privacy	**Phase 2**
NFR-12	Personal data shall not be shared with third parties without explicit consent.	0 third-party shares	GDPR / Privacy	**Phase 2**
NFR-13	The system shall not store any patient data or medical records.	0 patient records	GDPR / Privacy	**v1**
NFR-14	The system shall be accessible from Chrome, Firefox, Edge, and Safari (latest versions).	4 major browsers	Usability	**v1**
NFR-15	The application shall be responsive at 1280x720 to 1920x1080+.	Min 1280x720	Usability	**v1**
NFR-16	The UI shall follow WCAG 2.1 accessibility standards.	WCAG 2.1 Level AA	Accessibility	**Phase 2**
NFR-17	All interactive elements shall be keyboard-navigable.	100% keyboard-operable	Accessibility	**Phase 2**
NFR-18	Data shall be persisted with atomic write operations.	Atomic JSON writes	Reliability	**v1**
NFR-19	Activity logs shall be retained for audit.	>= 12 months retention	Reliability / Audit	**Phase 2**

5. Use Cases

The following use cases describe the primary interaction scenarios between users and the system. Each use case is marked with its implementation phase. Phase 2 use cases reflect the complete workflows defined in the project brief and are not yet implemented in the current codebase.

5.1 UC-01: Engineer/Healthcare Professional Registers with Institutional Email

Name	**UC-01: Register with Institutional Email** **[Phase 2]**
Actor(s)	Unregistered User (prospective Engineer or Healthcare Professional)
Precondition	The user is on the authentication screen and has navigated to the registration form.
Main Flow	1. The user clicks 'Create your membership' on the login page. 2. The system displays the registration form with fields for first name, last name, institutional email, phone, role (Engineer / Healthcare Professional), password, and password confirmation. 3. The user fills in all required fields and selects their role. 4. The user clicks 'Create account'. 5. The system validates that the email ends with .edu or .edu.tr [Phase 2]. 6. The system validates all other inputs (email format, phone length, password length, password match). 7. The system checks for duplicate email addresses. 8. The system hashes the password, creates an inactive account, and sends a verification email containing a unique confirmation link [Phase 2]. 9. The user opens the email and clicks the confirmation link; the account is activated. 10. The system redirects to the login form.
Postcondition	A new, activated account is created. The user can now log in with their institutional credentials.
Alternative Flow	5a. If the email is not institutional, the system shows 'Only .edu or .edu.tr email addresses are accepted.' [Phase 2]. 6a. If passwords do not match, the system shows 'The password confirmation does not match.' 7a. If the email already exists, the system shows 'An account with this email already exists.' 9a. If the verification link is not clicked within 24 hours, the account and pending registration are purged [Phase 2].

5.2 UC-02: User Logs In

Name	**UC-02: User Logs In** **[v1]**
Actor(s)	Registered User (Engineer / Healthcare Professional / Admin)
Precondition	The user has a valid, activated account.
Main Flow	1. The user enters their email and password on the login form. 2. The user clicks 'Enter Project Bridge'. 3. The system validates credentials using a timing-safe comparison. 4. The system verifies the account is active [Phase 2]. 5. The system returns a session object with userId, email, and role. 6. The session is stored in sessionStorage. 7. The system loads bootstrap data and redirects to the application shell. Admins see the Administrative Dashboard entry [Phase 2].
Postcondition	The user is authenticated and has access to features matching their role.
Alternative Flow	3a. If credentials are invalid, the system shows 'Incorrect email or password.' 4a. If the account is unverified, the system shows 'Please confirm your email before signing in.' [Phase 2].

5.3 UC-03: User Creates a Post (Draft or Active)

Name	**UC-03: User Creates a Post** [v1]
Actor(s)	Engineer or Healthcare Professional
Precondition	The user is logged in and has navigated to the 'Create' section.
Main Flow	1. The user clicks 'Create' in the navigation bar. 2. The system displays the post creation form with fields for title, working domain, required expertise, project stage, confidentiality level (Public / Confidential), city, and description. 3. The user fills in the form. 4. The user chooses 'Save as Draft' or 'Publish' [Phase 2]. 5. The system validates required fields. 6. The system creates the post with a unique ID, timestamps it, sets status to Draft or Active, and assigns the current user as owner. 7. If published, the post appears on the Project Board in Active status; if saved as draft, it appears only in the owner's 'My Posts' view [Phase 2].
Postcondition	The post is persisted with the appropriate lifecycle status and is visible to the correct audience.
Alternative Flow	5a. If required fields are missing, the system returns a validation error listing the missing fields. 4a. If the user closes the form without saving, v1 discards the entry; Phase 2 will offer autosave-as-draft.

5.4 UC-04: User Searches for Posts with Filters

Name	**UC-04: Search for Posts and Partners** [v1]
Actor(s)	Engineer or Healthcare Professional
Precondition	The user is logged in.
Main Flow	1. The user navigates to the Project Board or Network page. 2. The system displays all active posts and registered users. 3. The user applies one or more filters: Domain, City, Required Expertise, or Status [Phase 2 for status filter]. 4. The user may also enter a free-text search term. 5. The system filters the result set in real time, optionally prioritizing partners in the same city (city-based matching) [Phase 2]. 6. The user reviews the results.
Postcondition	The user sees a filtered list of posts or partners matching their criteria.
Alternative Flow	5a. If no results match, the system displays a friendly empty state with guidance.

5.5 UC-05: Interested Party Requests a Meeting (Full Workflow)

Name	**UC-05: Meeting Request Workflow** [Phase 2]
Actor(s)	Engineer or Healthcare Professional (as interested party), Post Owner
Precondition	The post is in Active status and the interested party is logged in.
Main Flow	1. The interested party opens the post detail page and clicks 'Express Interest' / 'Request Meeting'. 2. If the post's confidentiality level is Confidential, the system displays the NDA text and requires explicit acceptance; otherwise this step is skipped. 3. The interested party proposes one or more available time slots. 4. The post owner receives a notification and reviews the proposed slots. 5. The post owner accepts one slot. 6. The system transitions the post to 'Meeting Scheduled' state. 7. Both parties receive a confirmation notification

	with the confirmed time. The owner shares the external meeting link (Zoom/Teams) through the Focused Room. 8. The meeting takes place on the external platform. 9. After the meeting, the post owner may mark 'Partner Found' (terminal) or leave it Active to continue searching.
Postcondition	A meeting is confirmed, the post lifecycle state is updated, and an audit log entry is recorded.
Alternative Flow	2a. If the interested party declines the NDA, the request is cancelled. 5a. If no proposed slot is acceptable, the owner may counter-propose. 5b. If the owner does not respond within 7 days, the interest expression expires automatically.

5.6 UC-06: User Opens a Focused Room (v1 Interim)

Name	**UC-06: Open a Focused Room** [v1]
Actor(s)	Engineer or Healthcare Professional
Precondition	The user is logged in; at least one post and one other user exist.
Main Flow	1. The user clicks 'Start room' on a post or navigates to Focused Contact. 2. The system presents the room builder with post and collaborator selectors. 3. The user selects the post, the collaborator, and optionally sets a focus objective. 4. The user clicks 'Open focused room'. 5. The system checks for an existing room with the same participants and post; if one exists the user is redirected to it. 6. Otherwise, a new conversation is created with an auto-generated initial message. 7. The user is redirected to the Rooms Hub.
Postcondition	A Focused Room is created (or reused) and both participants can exchange messages attached to the post context.
Alternative Flow	5a. If a duplicate room exists, no new room is created and the existing room is opened instead. 4a. Attempting to create a room with oneself is rejected.

> **v1 IMPLEMENTATION NOTE** UC-06 is the v1 interim substitute for UC-05. In Phase 2, Focused Rooms become the conversation transcript attached to a confirmed meeting, not a standalone feature.

5.7 UC-07: Post Owner Marks Partner Found

Name	**UC-07: Mark Partner Found** [Phase 2]
Actor(s)	Post Owner (Engineer or Healthcare Professional)
Precondition	The owner has a post in Active or Meeting Scheduled state and has confirmed a suitable partner.
Main Flow	1. The owner opens their post detail page. 2. The owner clicks 'Mark as Partner Found'. 3. The system displays a confirmation dialog. 4. The owner confirms. 5. The system transitions the post to Partner Found state, rejects any pending interest expressions on the post, and writes an audit log entry [Phase 2].
Postcondition	The post is closed to new interest and appears with the Partner Found badge.
Alternative Flow	5a. The owner may reopen the post back to Active within 14 days if the partnership falls through [Phase 2].

5.8 UC-08: Admin Removes an Inappropriate Post

Name	UC-08: Admin Removes Inappropriate Post [Phase 2]
Actor(s)	Admin
Precondition	The Admin is logged in with Admin role privileges.
Main Flow	1. The Admin navigates to Administrative Dashboard → Posts. 2. The Admin filters posts by status, domain, or report flag. 3. The Admin opens a flagged or suspicious post. 4. The Admin reviews the content. 5. The Admin clicks 'Remove Post' and provides a reason (policy violation, spam, out-of-scope, etc.). 6. The system soft-deletes the post, notifies the owner, and writes an immutable audit log entry with the Admin's ID and reason.
Postcondition	The post is no longer visible to regular users; the audit trail records the removal.
Alternative Flow	6a. Soft-deleted posts remain in the database for audit purposes; only an Admin can restore them within 30 days.

5.9 UC-09: Admin Reviews Activity Logs

Name	UC-09: Admin Reviews Activity Logs [Phase 2]
Actor(s)	Admin
Precondition	The Admin is logged in with Admin role privileges.
Main Flow	1. The Admin navigates to Administrative Dashboard → Logs. 2. The Admin applies filters (date range, user ID, action type). 3. The system displays matching entries paginated (timestamp, user, action, target, IP, outcome). 4. The Admin clicks 'Export CSV' to download the filtered entries.
Postcondition	A CSV file with the filtered audit entries is downloaded.
Alternative Flow	3a. If no entries match, the system displays an empty-state message.

6. Data Model

The platform's data model spans both phases. v1 entities are implemented in the JSON-backed store. Phase 2 entities will be introduced alongside the meeting workflow, RBAC, and activity logging features.

6.1 User

Entity	Key Fields	Relationships
User [v1]	id (string, unique, prefix 'u'), name (string), role (enum: Engineer | Healthcare Professional | Admin [Phase 2]), specialty (string), city (string), status (string), bio (string), interests (string[]), availability (string), friendIds (string[]), isSuspended (bool) [Phase 2]	Has many Posts (as owner), Has many Conversations (via participantIds), Has many Connections (bidirectional via friendIds), Has one Account, Has many MeetingRequests [Phase 2], Has many ActivityLog entries [Phase 2]

6.2 Account

Entity	Key Fields	Relationships
Account [v1]	id (string, unique, prefix 'acc'), userId (FK), firstName, lastName, email (unique, normalized, .edu-validated [Phase 2]), phone, profession, passwordHash (scrypt salt:hash), emailVerified (bool) [Phase 2], verificationToken (string, nullable) [Phase 2], createdAt (ISO timestamp)	Belongs to one User

6.3 Post

Entity	Key Fields	Relationships
Post [v1]	id (string, unique, prefix 'p'), title, domain, summary, description [Phase 2], requiredExpertise [Phase 2], projectStage (enum), confidentialityLevel (enum: Public|Confidential) [Phase 2], city [Phase 2], status (enum: Draft|Active|Meeting Scheduled|Partner Found|Expired) [Phase 2], ownerId (FK), tags (string[]), createdAt, updatedAt [Phase 2], expiresAt [Phase 2]	Belongs to one User (owner), Has many MeetingRequests [Phase 2], Has many Conversations

6.4 MeetingRequest

Entity	Key Fields	Relationships
MeetingRequest [Phase 2]	id (string, unique, prefix 'mr'), postId (FK), requesterId (FK), ownerId (FK), status (enum: Pending|NDA Accepted|Slots Proposed|Confirmed|Declined|Expired), ndaAcceptedAt (nullable), proposedSlots (ISO timestamp[]), confirmedSlot (nullable),	Belongs to one Post, Belongs to one User (requester), Belongs to one User (owner), Has one Conversation (after confirmation)

	externalMeetingLink (nullable), createdAt, resolvedAt (nullable)	

6.5 Conversation (Focused Room)

Entity	Key Fields	Relationships
Conversation [v1]	id (string, unique, prefix 'c'), title (auto-generated), projectId (FK to Post), meetingRequestId (FK, nullable) [Phase 2], participantIds (FK[], sorted pair), focus (string), messages (Message[])	Belongs to one Post, Belongs to two Users (participants), Belongs to one MeetingRequest [Phase 2]

6.6 Message

Entity	Key Fields	Relationships
Message [v1]	id (string, unique, prefix 'm'), senderId (FK), body (string), createdAt (ISO timestamp)	Belongs to one Conversation (embedded), Belongs to one User (sender)

6.7 ActivityLog

Entity	Key Fields
ActivityLog [Phase 2]	id (string, unique, prefix 'log'), userId (FK, nullable), actionType (enum: REGISTER|LOGIN|POST_CREATE|POST_EDIT|POST_DELETE|INTEREST_EXPRESS|NDA_ACCE targetEntityType, targetEntityId, ipAddress, outcome (enum: SUCCESS|FAILURE), details (JSON, null

7. Interface Requirements

7.1 User Interface (UI)

The platform consists of a two-phase user interface: an authentication screen and the main application shell. The main shell exposes navigation sections tailored to the user's role.

7.1.1 Authentication Screen

The authentication screen contains two views: Login and Registration. The login form includes email and password fields. The registration form captures first name, last name, institutional email [Phase 2], phone, role selection (Engineer / Healthcare Professional) [Phase 2], password, and password confirmation. In v1, the role field accepts a broader list of profession options and does not yet enforce role-based permissions. Both forms display contextual notices with auto-dismiss behaviour.

7.1.2 Main Application Shell (Regular Users)

After login, the application presents a top bar with brand name, dynamic page heading, theme toggle, and sign-out button. A horizontal section navigation bar provides access to: Overview (hero and statistics), Project Board (post list with filters), Network (searchable people cards), Create (post creation form), Focused Contact (room builder, v1) / Meeting Request (Phase 2), Rooms Hub (conversation threads), and Member Profiles (profile viewer). In Phase 2, posts will display a colored status badge (Gray/Green/Blue/Purple/Red) reflecting their lifecycle state.

7.1.3 Administrative Dashboard [Phase 2]

Admin-role users see an additional 'Admin' navigation entry leading to the Administrative Dashboard, with sub-sections for Posts (browse/filter/remove), Users (browse/filter/suspend), Logs (filter/export), and Statistics (platform-wide KPIs: users per role, posts per status, meetings scheduled per month, partners found per month).

7.1.4 Visual Theming

The platform supports three color themes cycleable via a toggle button: Neon Pulse (default), Earth Canvas, and Citrus Tide. Theme selection is persisted during the session.

7.2 API Endpoints

Method	Endpoint	Description	Phase
GET	/api/bootstrap	Returns users, projects, conversations, and platform stats	v1
POST	/api/register	Creates a new account and user profile	v1
POST	/api/login	Authenticates user and returns session	v1
POST	/api/verify-email	Confirms an email address via token	Phase 2
POST	/api/password-reset	Triggers a password reset email	Phase 2
POST	/api/projects	Creates a new post	v1
PATCH	/api/projects/:id	Edits an existing post (owner only)	Phase 2
PATCH	/api/projects/:id/status	Transitions post status (Draft→Active, →Partner Found, etc.)	Phase 2
DELETE	/api/projects/:id	Soft-deletes a post (admin only)	Phase 2
POST	/api/friends	Creates a bidirectional connection	v1

POST	/api/conversations	Creates a new Focused Room	v1
POST	/api/messages	Sends a message in a room	v1
POST	/api/meeting-requests	Creates a meeting request with optional NDA acceptance	Phase 2
PATCH	/api/meeting-requests/:id	Propose slots / accept slot / decline	Phase 2
GET	/api/admin/logs	Returns filtered activity log entries (admin only)	Phase 2
GET	/api/admin/logs/export	Exports log entries as CSV (admin only)	Phase 2
PATCH	/api/admin/users/:id	Suspends or reactivates a user (admin only)	Phase 2

7.3 External System Interfaces

The v1 implementation has no external system dependencies; the server uses only Node.js built-in modules. Phase 2 introduces dependencies on (a) a transactional email service (e.g., SendGrid, Amazon SES) for email verification, password reset, and meeting notifications, and (b) a relational database (e.g., PostgreSQL) to support activity logging, meeting workflows, and higher concurrency. Meetings themselves continue to be conducted on external platforms (Zoom, Microsoft Teams) with only the meeting link shared through the Focused Room.

8. Requirements Traceability Matrix

The following matrix maps each functional requirement to its source in the project brief (or implementing code component) and related use case(s).

Req. ID	Requirement Summary	Source (Brief / Code)	Related Use Case	Phase
FR-01	.edu email restriction	Brief 4.1	UC-01	Phase 2
FR-02	Email verification mechanism	Brief 4.1	UC-01	Phase 2
FR-03	Role selection at registration	Brief 4.1	UC-01, UC-02	Phase 2
FR-04	Registration with full user details	server.js /api/register	UC-01	v1
FR-10	Login with email/password	server.js /api/login	UC-02	v1
FR-14	Role-Based Access Control	Brief 4.5	UC-02, UC-08, UC-09	Phase 2
FR-20	Post creation	server.js /api/projects	UC-03	v1
FR-22	Post lifecycle states	Brief 4.2	UC-03, UC-05, UC-07	Phase 2
FR-25	Mark Partner Found transition	Brief 4.2	UC-07	Phase 2
FR-31	Confidentiality level on posts	Brief 4.2	UC-03, UC-05	Phase 2
FR-41	Filter posts by domain/expertise/status/city	Brief 4.3	UC-04	Phase 2
FR-42	City-based matching	Brief 4.3	UC-04	Phase 2
FR-50	Express interest / request meeting	Brief 4.4	UC-05	Phase 2
FR-51	NDA acceptance on Confidential posts	Brief 4.4	UC-05	Phase 2
FR-53	Propose meeting time slots	Brief 4.4	UC-05	Phase 2
FR-54	Accept proposed slot	Brief 4.4	UC-05	Phase 2
FR-58	Focused Room (v1 interim)	server.js /api/conversations	UC-06	v1
FR-70	Administrative Dashboard	Brief 4.5	UC-08, UC-09	Phase 2
FR-71	Admin post removal	Brief 4.5	UC-08	Phase 2
FR-73	Suspend user accounts	Brief 4.5	UC-09	Phase 2
FR-80	Activity logging	Brief 4.6	UC-05, UC-07, UC-08, UC-09	Phase 2
FR-82	Immutable log entries	Brief 4.6	UC-09	Phase 2
FR-92	GDPR account deletion	UserGuide 6.2	—	Phase 2
FR-93	GDPR data export	UserGuide 6.3	—	Phase 2
FR-105	In-app notifications	UserGuide 8	UC-05, UC-07	Phase 2
FR-106	Email notifications	UserGuide 8	UC-01, UC-05	Phase 2

9. Technical Implementation Details

9.1 Technology Stack

Layer	Technology (v1)	Technology (Phase 2)
Frontend	Vanilla JavaScript (ES6+), HTML5, CSS3	Same, with additional components for Admin Dashboard and meeting workflow UI
Backend	Node.js (built-in modules: http, path, fs/promises, crypto)	Node.js + Express/Fastify, scheduled jobs for post expiration
Data Storage	JSON file (db.json) with atomic write queue	PostgreSQL (or equivalent) with migrations
Authentication	scrypt + sessionStorage	scrypt + httpOnly cookies + JWT refresh tokens + email verification
Email	Not used	Transactional email service (SendGrid / Amazon SES)
Deployment	Vercel serverless or local Node	Vercel + external DB, or containerized deployment
Audit	Not implemented	Append-only audit log with 12-month retention

9.2 Security Measures (v1)

The v1 platform implements: scrypt password hashing with 16-byte salt and 64-byte key; timing-safe comparison via crypto.timingSafeEqual; input sanitization through cleanText(), normalizeEmail(), and normalizePhone(); XSS prevention via escapeHtml(); request body size limit of 1 MB; case-insensitive duplicate detection via email normalization. Phase 2 adds HTTPS enforcement, session token expiration, institutional email validation, email verification, and an immutable audit trail.

9.3 Data Persistence Strategy

The v1 persistence layer uses a JSON file with a sequential write queue. The readDb() function reads and normalizes the JSON; the updateDb() function chains write operations through a Promise queue (writeQueue), ensuring that concurrent mutations are processed sequentially to prevent race conditions. On Vercel, the database file is copied to /tmp on first access; this storage is ephemeral and is not suitable for production. Phase 2 migrates to a managed relational database with connection pooling, transactions, and proper backup/restore procedures.

9.4 Project File Structure (v1)

File / Directory	Purpose
server.js	HTTP server, API route handlers, authentication, input validation
lib/store.js	Data access layer with readDb(), updateDb(), and write queue
api/index.js	Vercel serverless function adapter
public/index.html	Main HTML with auth screens and app shell sections
public/app.js	Client-side SPA: state, rendering, events, API calls

public/styles.css	CSS with theme support and responsive layout
data/db.json	JSON data store: users, accounts, projects, conversations
vercel.json	Vercel deployment configuration
package.json	Project metadata and start script

10. Appendices

10.1 Demo Accounts (v1)

The system includes five pre-seeded demo accounts for testing and demonstration. All demo accounts use the password 'bridge123'. In Phase 2, demo accounts will be migrated to .edu addresses.

Name	Email	Role	Specialty	City
Dr. Selin Kaya	selin@projectbridge.demo	Healthcare Professional (Doctor)	Cardiology	Istanbul
Dr. Emre Demir	emre@projectbridge.demo	Healthcare Professional (Doctor)	Radiology	Ankara
Mert Aydin	mert@projectbridge.demo	Engineer	Embedded Systems	Izmir
Leyla Arslan	leyla@projectbridge.demo	Engineer	AI and Data Systems	Berlin
Aylin Cakir	aylin@projectbridge.demo	Engineer	Frontend Product Design	Amsterdam

10.2 Sample Posts (v1)

Post Title	Domain	Project Stage	Owner Role	Owner
Smart Rehab Glove	Rehabilitation Tech	Prototype	Engineer	Mert Aydin
Radiology Triage Console	Clinical AI	Discovery	Healthcare Professional	Dr. Emre Demir
Heart Failure Companion	Digital Health	Validation	Healthcare Professional	Dr. Selin Kaya

10.3 Phase 2 Delivery Summary

The following capabilities are planned for Phase 2 and are tracked throughout this document with [Phase 2] tags:

- Institutional email (.edu / .edu.tr) restriction and email verification at registration
- Three-role model (Engineer, Healthcare Professional, Admin) with Role-Based Access Control
- Post lifecycle states: Draft, Active, Meeting Scheduled, Partner Found, Expired
- Confidentiality level on posts (Public / Confidential) and NDA acceptance workflow
- Meeting request workflow: interest expression, slot proposal, owner acceptance, confirmation
- Administrative Dashboard with post moderation, user management, and platform statistics
- Immutable activity log with filtering and CSV export for audit and compliance
- City-based partner matching and status-based post filtering
- GDPR compliance: account deletion (right to erasure) and data export (right to portability)

- Notification system: in-app and email (interest received, meeting confirmed, account activity)
- Password reset via email link
- Migration to a relational database for concurrent access and audit retention

10.4 Appendix Attachments

- Use Case Diagram (to be attached)
- Wireframes / Screen mockups (to be attached)
- Entity-Relationship Diagram (to be attached)
- Post lifecycle state-transition diagram (to be attached)
- Meeting request sequence diagram (to be attached)